|INCH-POUND|
MIL-R-5001B
31 July 1992
SUPERSEDING
MIL-R-5001A
14 October 1965

MILITARY SPECIFICATION

RUBBER CELLULAR SHEET, MOLDED AND HAND-BUILT SHAPES; LATEX FOAM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cored and uncored open cell latex foam rubber sheets, molded and hand-built shapes having good shock-absorbing, vibration-damping and comfort cushioning characteristics (see 6.1).

1.2 Classification.

1.2.1 Types. Latex foam rubber covered by this specification shall be furnished in the following types as specified (see 6.2):

Type I - Cored
Type II - Uncored

1.2.1.1 Composition. Latex foam rubber covered by this specification shall be furnished in the following compositions as specified (see 6.2):

Composition 1 - Lattices of natural rubber, synthetic natural rubber (polyisoprene) or a combination thereof.
Composition 2 - Lattices of synthetic rubber (non-polyisoprene), or a combination of synthetic natural rubber (polyisoprene) with synthetic rubber.

1.2.2 Classes.

1.2.2.1 Type I. Type I latex foam rubber covered by this specification shall be furnished in the following classes as specified (see 6.2):

Class 1 - Soft
Class 2 - Medium
Class 3 - Firm

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Warfare Center, Aircraft Division Lakehurst, Code SR3, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9320
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

1.2.2.2 Type II. Type II latex foam rubber covered by this specification shall be furnished in the following classes as specified (see 6.2):

Class 1 - Soft
Class 2 - Medium
Class 3 - Firm
Class 4 - Extra Firm

1.2.3 Grades. Latex foam rubber covered by this specification shall be furnished in the following grades as specified (see 6.2):

Grade A - Flame Resistant.
Grade B - Nonflame Resistant, moderate low temperature compression.
Grade C - Nonflame Resistant, good low temperature compression.

2. APPLICABLE DOCUMENTS

2 1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Box, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U. S. Military Property
MIL-STD-293 - Visual Inspection Guide for Cellular Rubber Items

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D1055 - Flexible Cellular Material - Latex Foam.
ASTM D4986 - Horizontal Burning Characteristics of Cellular Polymeric Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Material. The latex foam shall be made from vulcanized rubber lattices of Composition 1 or Composition 2 material as specified (see 6.2). The cells shall be open and interconnecting having a uniform structure. Except for the allowances specified in 3.6, there shall be no large seams in the material and the surfaces of the material shall be porous.

3.2.1 Limitation on age. Material shall not be older than one year from the date of manufacture at the time of shipment to the Government.

3.3 Construction.

3.3.1 Form. Latex foam shall be furnished in the form of sheets, molded shapes, or hand-built shapes as specified (see 6.2). Assembled products may be provided when the finished product desired is larger than it is standard practice in industry to mold in one section.

3.3.2 Cores. Size, shape and distribution of coring shall be at the manufacturer's option, unless otherwise specified by the acquiring activity (see 6.2).

3.4 Surfacing. Latex foam shall be backed or surfaced with fabric, cement or other material as specified in the contract or purchase order, or by applicable drawings (see 6.2).

3.5 Finish.

3.5.1 Sheets. Unless otherwise specified (see 6.2), sheets shall be Type II and smooth on both faces. Fabric impressions are not objectionable.

3.5.2 Molded or hand-built shapes. Unless otherwise specified (see 6.2), molded or hand-built shapes shall be smooth on all surfaces.

3.6 Allowable manufacturing defects. The following defects may be allowed in latex foam when their occurrence is incident to standard manufacturing practice and assembly, providing they are corrected as indicated herein and the resulting product conforms with the requirements of this specification and supplementary specifications and drawings:

Source: https://assist.dla.mil Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

a. Tears occurring during removal from mold which have been completely cemented together.
b. Changes in the volume of product during cure resulting in shallow surface depressions or minor changes in size (not exceeding 1 inch in width) which have been corrected by addition of material of identical composition and quality as that of the product and by buffing to a level, neat-appearing surface.
c. Low-compression cushions adjusted by the addition of material of identical composition and quality in the core holes in a symmetrical pattern throughout the product, to produce the required uniform compression resistance.

3.7 Tolerances. Unless otherwise specified in the contract or purchase order (see 6.2), tolerances on dimensions of sheet material shall be as specified in Table I. Tolerances on dimensions of molded or hand-built shapes shall be as specified by the applicable drawings or specifications.

TABLE I. Tolerances on dimensions of sheet material.

Type	Dimension	Range (inches)	Tolerances (inches)	
			Plus	Minus
I	Length and width	0 - 6	.188	.063
		6 - 12	.375	.125
		12 - 24	.500	.250
		24 - 36	.625	.375
		36 - 48	.750	.500
		48 - 60	.875	.625
		60 - 72	1.000	.750
		72 and over	1.125	.875
	Thickness	0 - 3	.125	.063
		3 - 5	.188	.125
		5 and over	.250	.188
II	Length and width	0 - 6	.313	.063
		6 - 12	.500	.125
		12 - 24	.688	.250
		24 - 36	.875	.375
		36 - 48	1.125	.500
		48 - 60	1.250	.625
		60 - 72	1.375	.750
		72 and over	1.500	.875
	Nominal thickness	Up to and including 1/2	.063	.063
		Over 1/2 to 1 inclusive	.125	.063
		Over 1 to 3 inclusive	.188	.125
		3 and over	.250	.188

3.8 Odor. The latex foam shall be free from any objectionable odor. When specified (see 6.2), the material shall be treated to minimize the odor.

3.9 Indentation. The force required to produce a 25 percent indentation shall be as specified in Table II for each class of Type I (cored) material and Table III for each class of Type II (uncored) material. Testing shall be conducted in accordance with 4.5.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

TABLE II. Indentation resistance and apparent density for Type I material.

CLASS	INDENTATION RESISTANCE (pounds force to produce a 25% indentation over a 50 in^2 area)	APPARENT DENSITY 1/ (lb/cu. in.) (Maximum)													
		GRADE A Thickness (inches)							GRADE B AND C Thickness (inches)						
		1	1.50	2	2.50	3	3.50	Over 3.50	1	1.50	2	2.50	3	3.50	Over 3.50
Soft	10 to 20	.0050	.0047	.0044	.0041	.0039	.0038	.0037	.0039	.0036	.0033	.0030	.0028	.0027	.0026
Medium	Over 20 to 35	.0061	.0057	.0052	.0048	.0046	.0045	.0044	.0050	.0046	.0041	.0037	.0035	.0034	.0033
Firm	Over 35 to 55	.0069	.0065	.0060	.0056	.0054	.0052	.0050	.0058	.0054	.0049	.0045	.0042	.0040	.0038

1/ Apparent density is based on gross volume and includes coring.

TABLE III. Indentation resistance and density for Type II material.

CLASS	INDENTATION RESISTANCE (Pounds force to produce a 25% indentation over a 50 in^2 area)	DENSITY (lb/cu. in.) (Maximum)	
		GRADE A	GRADE B AND C
Soft	10 to 25	0.0060	0.0045
Medium	Over 25 to 40	0.0085	0.0055
Firm	Over 40 to 60	0.0100	0.0065
Extra Firm	Over 60 to 85	0.0120	0.0075

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

3.10 Low temperature compression deflection (percent change). The maximum percent change in compression deflection at -40°F (-40°C) for each material grade shall be as follows when tested in accordance with 4.5.2:

Grade A - 70%
Grade B - 60%
Grade C - 25%

3.11 Density. The apparent density for each class of Type I material and the density for each class of Type II material shall be as specified in Table II and III respectively when tested in accordance with 4.5.3.

3.12 Flame resistance (Grade A only). Grade A material shall not support a flame for longer than 50 seconds when tested in accordance with 4.5.4.

3.13 Compression set. The compression set shall not be greater than 10 percent (expressed as a percentage of the original height) when tested in accordance with 4.5.5.

3.14 Flexing endurance. After 250,000 compression cycles, there shall be no breakdown of the cellular structure and the compression set shall not be greater than 10 percent when tested in accordance with 4.5.6.

3.15 Resistance to accelerated aging. After accelerated aging, the change in compression deflection shall not be greater than 15 percent when tested in accordance with 4.5.7.

3.16 Color. Unless otherwise specified (see 6.2), the color of the latex foam rubber shall be "as manufactured."

3.17 Identification marking In addition to any special marking required by the contract or order (see 6.2), each piece of latex foam rubber shall be marked in a permanent and legible manner in accordance with the marking requirements of MIL-STD-130 and shall also include the Type, Class, Grade, and Date of manufacture. The identification marking shall be stamped or printed with indelible ink which is harmless to the materials.

3.17.1 Sheets. The identification marking on sheet material shall be applied on one side only in rows spaced not more than 5 inches apart of constantly recurring characters not less than .375 inch in height extending either lengthwise or crosswise of the sheet.

3.18 Workmanship. The latex foam shall be of uniform texture and shall be free from manufacturing imperfections which may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil - Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of all the tests in Table IV and the examination for defects in appearance and construction as specified in 4.4.2 for each type, class and grade of latex foam rubber submitted for approval. Prior to contract award, bidders shall provide the procuring activity with a Material Safety Data Sheet (MSDS) prepared in accordance with FED-STD-313. The responsibility for the performance of the first article inspection shall be as specified in the contract or purchase order (see 6.2, 6.2.2, and 6.4). All material supplied to this specification shall be manufactured using the same formulation and methods used to obtain the first article samples unless a change has been approved by the procuring activity.

TABLE IV. First article tests.

Test	Requirement Paragraph	Test Method	Number of Tests Required 1/	Report Average Result Numerically To the Nearest
Indentation Resistance	3.9	4.5.1	3	1 Lbf
Low Temp Compression Deflection	3.10	4.5.2	3	0.1 percent
Density	3.11	4.5.3	3	0.0001 lbs./in.^{3}
Flame Resistance Grade A Only	3.12	4.5.4	3	1.0 second
Compression Set	3.13	4.5.5	3	0.1 percent
Flexing Endurance	3.14	4.5.6	3	0.1 percent
Accelerated Aging	3.15	4.5.7	3	0.1 percent

1/ - Test reports shall include individual values on which average results are based.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

4.3.1 Sampling for first article inspection. As soon as possible after the award of a contract or purchase order, the contractor shall submit at least three 12 by 12 by 1 inch pieces of Type II material in the grade and class specified in the contract or purchase order for first article inspection. Approval of Type I material will be granted upon approval of samples of Type II material of the same composition. Further production of latex foam rubber by the contractor prior to the approval of the acquiring activity or completion of inspection of the first article shall be at the contractor's risk.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of inspection for appearance and construction (see 4.4.2), dimensional inspection (see 4.4.3), packaging inspection (see 4.4.4), and physical and mechanical property testing (see 4.4.5). All quality conformance inspections and tests shall be conducted on each lot of material (see 4.4.1). In addition, the contractor shall supply a Material Safety Data Sheet (MSDS), prepared in accordance with FED-STD-313, with each shipment of material from a lot.

4.4.1 Inspection lot. All material of the same type, composition, class, grade and size (or part number) manufactured under the same conditions from the same batch or batches of latex foam rubber compound and offered for delivery at one time shall be considered a lot for purposes of inspection and tests.

4.4.2 Inspection for appearance and construction. Sample units for inspection of appearance and construction shall be selected in accordance with Inspection Level II of MIL-STD-105. The sample unit for this examination shall be one sheet, one molded shape, or one hand-built shape, as applicable. Inspection shall be performed in accordance with Table V. Acceptance criteria shall be as specified in the contract or purchase order (see 6.2 and 6.6).

TABLE V. Inspection for appearance and construction.

Examine	Defect	Major	Minor
Appearance	Not uniform in texture, finish or firmness	X	
	Presence of dirt, foreign material, embedded particles		X
Construction	Any defects classified as major or minor defects for latex foam as specified in MIL-STD-293		
	Surface or backing not as specified	X	

4.4.3 Dimensional inspection. Sample units for dimensional inspection shall be selected in accordance with Inspection Level II of MIL-STD-105. The sample unit for this examination shall be one sheet, one molded shape, or one hand-built shape, as applicable. Inspection shall be performed in accordance with Table VI. Acceptance criteria shall be as specified in the contract or purchase order (see 6.2 and 6.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

TABLE VI. Dimensional inspection.

Examine	Defect
Sheets	Any dimension that varies by more than plus or minus the applicable tolerances specified in 3.7, in drawings, or in the contract.
Molded parts	Any dimension that varies by more than plus or minus the specified size or tolerance allowed in the applicable drawings or contract.

4.4.4 Packaging inspection. Packaging inspection shall be performed to determine that preservation, packing, and markings comply with the requirements of section 5 of this specification. Sample units for packaging inspection shall be selected in accordance with Inspection Level S-2 of MIL-STD-105. The sample unit for this examination shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects. Inspection shall be conducted in accordance with Table VII. Acceptance criteria shall be as specified in the contract or purchase order (see 6.2 and 6.6).

TABLE VII. Packaging inspection.

Examine	Defect
Preservation	Molded shapes not wrapped, boxed or otherwise protected against abrasion and deformation. Unit package not as specified. Packaging material not as specified.
Packing	Not in accordance with contract requirements. Arrangement or number of units per shipping container not in accordance with contract requirements. Container not as specified or required methods of materials. Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability. Inadequate application of components, such as: incomplete closures of case liners, container flaps, loose or inadequate strapping, bulged or distorted containers.
Count	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings (as applicable) omitted, illegible, incorrect, incomplete or not in accordance with contract requirements. Precautionary markings missing or not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

4.4.5 Physical and mechanical property testing. A sufficient number of items to conduct the physical and mechanical property tests specified in Table VIII shall be randomly selected from each lot of material (see 4.4.1). Except as specified below, the required test specimens shall be fabricated from the supplied sheet, molded parts, or hand-built parts, as applicable. No more than one determination per test shall be conducted on each item selected from the lot. For example, conducting two of the three compression set tests on one of the items selected shall not be permitted. If the items supplied are of such shape that test specimens can not be obtained from them, they shall be fabricated from sheets having a length of 12 inches, a width of 12 inches, and a thickness of 1 inch. The sheets shall be identical in cure and composition as the lot of material they represent and no more than one determination per test shall be conducted on each sheet.

TABLE VIII. Physical and mechanical property testing.

Test	Requirement Paragraph	Test Method	Number of Tests Required 1/	Report Average Result Numerically To the Nearest
Indentation Resistance	3.9	4.5.1	3	1 Lb_f
Density	3.11	4.5.3	3	0.0001 lbs./in.^{3}
Flame Resistance Grade A Only	3.12	4.5.4	3	1.0 second
Compression Set	3.13	4.5.5	3	0.1 percent

1/ - Test reports shall include individual values on which results are based.

4.5 Test methods. All tests shall be conducted as specified herein. Tests shall be conducted at a temperature of 75° ± 5°F (24° ± 3°C) unless otherwise specified in the applicable test method.

4.5.1 Indentation test. The indentation test shall be conducted in accordance with the method specified in ASTM D1055 (see 6.5). Testing shall be performed on 3 specimens and the average value used to determine conformance with 3.9.

4.5.2 Low temperature compression deflection test (percent change). Testing to determine the percent change in compression deflection after exposure to -40°F (-40°C) shall be conducted in accordance with the low temperature test method (suffix F1) specified in ASTM D1055 (see 6.5). Testing shall be performed on 3 specimens and the average percent change in compression deflection used to determine conformance with 3.10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

4.5.3 Density. The density shall be the average of three determinations. Each determination shall consist of accurately weighing and measuring a sample and dividing the weight by the volume. Samples to be tested shall have been placed in a desiccator for 24 hours immediately prior to density determination. The volume of irregularly shaped products or specimens, which cannot be measured by ordinary means, shall be obtained as follows: If the product has a cored side, the core holes shall be sealed off with cellophane or paper sheeting. The sample shall be placed in a graduated container of slightly larger dimensions than the sample and completely surrounded by 10 to 14 mesh, free flowing seed (such as flaxseed) so that the sample is covered to a depth of 1 inch. The seed shall be added to the graduated container from a point approximately 6 inches above the top of the product. A glass powder funnel having a top diameter of 100 mm and a stem diameter of 18 mm is recommended for adding the flaxseed. The true or apparent volume (as applicable) of the seed plus the sample shall be recorded, the sample removed and the volume of the seed determined. The volume (or apparent volume) of the latex foam shall be determined by deducting the volume of the seed from the volume of the seed plus the sample. The density (or apparent density in the case of cored material) shall be determined by dividing the weight by the volume.

4.5.4 Flame resistance. The time the material supports a flame shall be determined in accordance with ASTM D4986 except as specified herein. Testing shall be performed on three specimens 0.250 inches thick by 0.500 inches wide and of any convenient length. Prior to testing, specimens shall be conditioned for a minimum of 48 hours at 73 ± 4°F (23 ± 2°C) and a relative humidity of 50 ± 5 percent. The specimens shall be placed on the support gauze holder such that the thickness by length area is horizontal. The average time of flame propagation shall be used to determine conformance to 3.12.

4.5.5 Compression set. Compression set, expressed as a percentage of the original height, shall be determined in accordance with the method specified in ASTM D1055 (see 6.5). Testing shall be performed on 3 specimens and the average value used to determine conformance with 3.13.

4.5.6 Flexing endurance. Three specimens shall be tested for flexing endurance in accordance with the flexing method specified in ASTM D1055 (see 6.5). After 250,000 cycles, the cellular structure of each specimen shall be visually examined for conformance to 3.14. In addition, the average value for compression set shall meet the requirements in 3.14 when determined as follows:

$$\text{Compression set} = (t_o - t_1)/t_o \times 100$$

Where:

t_o = Original specimen height
t_1 = Specimen height one hour after completion of the 250,000 cycles

4.5.7 Resistance to accelerated aging. Three specimens, identical to those required in the low temperature compression deflection test in 4.5.2, shall be conditioned for accelerated aging in accordance with the air-oven method described in ASTM D1055 (see 6.5). Specimens shall be subjected to 212° ± 2°F (100° ± 1°C) for a period of 70 ± 1/2 hours. After conditioning, the test specimen shall be allowed to rest undeflected and undistorted at 75° ± 5°F (24° ± 3°C) for at least 22 hours before testing for percent change in compression deflection as specified in ASTM D1055. The average value shall be used to determine conformance to 3.15.

Source: https://assist.dla.mil Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Molded shapes shall be wrapped, boxed, or otherwise protected against deformation and abrasion. Latex foam rubber sheets shall be preserved in accordance with supplier's commercial practice.

5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.2).

5.2.1 Level A. The latex foam rubber preserved in accordance with 5.1 shall be packed in snug fitting fiberboard boxes conforming to PPP-B-640, Class 2, Grade A, or in overseas type cleated-plywood boxes conforming to PPP-B-601 (see 6.2). Boxes shall be strapped in accordance with the appendix of the applicable box specification.

5.2 2 Level B. The latex foam rubber preserved in accordance with 5.1 shall be packed in snug fitting fiberboard boxes conforming to PPP-B-636, Class - domestic, PPP-B-640, Class 1, Grade B, or in domestic type cleated boxes conforming to PPP-B-601 (see 6.2).

5 2.3 Level C. The latex foam rubber preserved in accordance with 5.1 shall be packed in a manner to ensure carrier acceptance and safe delivery at lowest transportation cost to destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and shall include the following information:

Stock No. or other identification number as specified in the purchase document

RUBBER, CELLULAR SHEET, MOLDED AND HAND-BUILT SHAPES; LATEX FOAM

MIL-R-5001A

Type_____ Composition_____ Class_____ Grade_____

Manufacturer's Designation No.

Date of manufacture

Name of manufacturer

5.3.1 Precautionary marking. The following precautionary marking shall be included on each exterior shipping container.

"STORE IN A COOL PLACE"

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

6.1 Intended use. The cellular latex foam sheet, molded and hand-built shapes covered by this specification is intended for use in crash protection pads, parachute seat cushions and back pads, goggle cushions, and other uses requiring a material capable of absorbing shock and damping vibration and having a relatively low-compression resistance.

6.1.1 Grade. The intended use for grades of Type I (cored) and for Type II (uncored) are as follows:

a.	Grade A:	Grade A material is used when resistance to flame is of primary importance.
b.	Grades B and C:	Grades B and C materials are used when lighter weight and low temperatures are of primary importance.

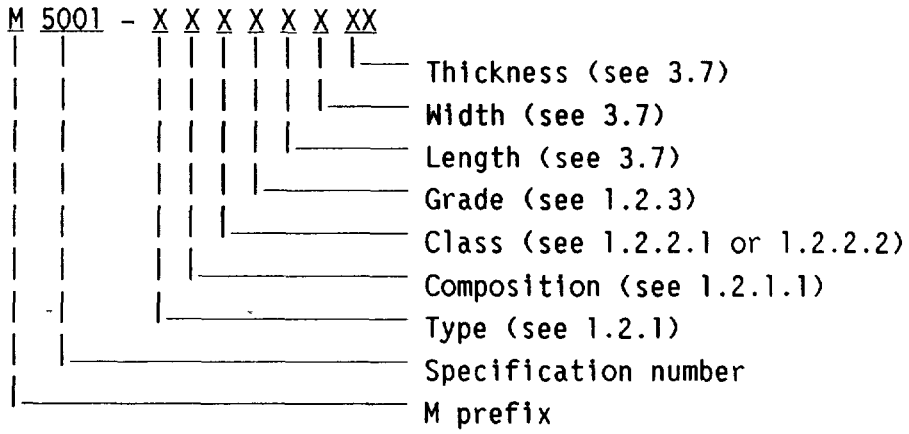
6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Number of pieces required.
c. Detail drawings or specifications describing the shape and surfacing of the part.
d. Type, Composition, Class and Grade required (see 1.2 and 6.1).
e. Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
f. If first article testing is not required (see 3.1).
g. Form required (see 3.3.1).
h. Size, shape and distribution of coring required (see 3.3.2).
i. When surfacing is required (see 3.4).
j. Finish required (see 3.5).
k. Dimensions (for sheet material) (see 3.7) and tolerances if different than that in Table I.
l. When treatment is required to minimize odor (see 3.8).
m. Color, other than "as manufactured" required (see 3.16).
n. Responsibility for the performance of the first article inspection (see 4.3).
o. Acceptance criteria (see 6.6) for appearance and construction inspection (see 4.4.2), dimensional inspection (see 4.4.3), and packaging inspection (see 4.4)
p. Level of packing required (see 5.2).
q. Special marking requirements (see 3.17 and 5.3).

6.2.1 Part or identifying number (PIN) for sheet material. Part numbers should be assigned to identify sheet material in accordance with this specification. Military part numbers should be coded in accordance with the following example:

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

PIN EXAMPLE:



6.2.2 Consideration of data requirements. The following data requirement should be considered when this specification is applied on a contract The applicable Data Item Description (DID) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID is tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27-4.75-1 exempts the requirement for a DD Form 1423.

Reference Para.	DID Number	DID Title	Suggested Tailoring
4.1.1 4.3	DI-NDTI-80809A	TEST/INSPECTION REPORTS	

The above DID was cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.3 Design information.

6.3.1 Material classes. The division into classes specified herein is based on the assumption that working loads will compress (indent) the latex foam rubber to 75 percent of its normal or unloaded thickness. Compression resistance, as defined herein, is therefore a measure of the normal working load which should be applied to any class of material.

6.3.2 Material type. If a particular type of latex foam rubber is desired, it should be specified; otherwise, the manufacturer may supply either Type I or Type II. The approximate compression resistance and density should be determined and detail drawings and specifications should designate the material accordingly. See Tables II and III.

6.3.3 Polyurethane foam. This specification does not cover polyurethane foam. Designers requiring flexible polyurethane foam should consider using L-P-386 or current industry documents approved as listed in the Department of Defense Index of Specifications and Standards.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

6.4 First article. The contracting officer should provide specific guidance to offerors whether the first article samples should be a preproduction item, a first production item, a sample selected from the first production items, or a standard production item from the contractors inventory (see 3.1 and 4.3). The contracting officer should also include specific instructions in acquisition documents regarding arrangements for submission of Material Safety Data Sheets (MSDS) prior to contract award, testing, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive requirements for first article inspection to those bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Test sections in ASTM D1055. As test sections may change upon subsequent revisions to ASTM D1055, Table IX outlines the applicable sections of ASTM D1055-90, which was the most current at the time of this revision, for the tests required in this document.

TABLE IX. Test sections in ASTM D1055.

Test	Applicable Sections of ASTM D1055	Requirement Paragraph	Test Paragraph
Indentation Resistance	20 - 23	3.9	4.5.1
Low Temp Compression Deflection	27 - 30	3.10	4.5.2
Compression Set	17 - 19	3.13	4.5.5
Flexing Endurance	24 - 26	3.14	4.5.6
Accelerated Aging 1/	15 - 16	3.15	4.5.7

1/ - Percentage change in compression-deflection is determined using the method outlined in sections 27 - 30 of ASTM D1055 except the results are based on percent change before and after accelerated aging.

6.6 Acceptance criteria. The acceptance criteria for appearance and construction, dimensional, and packaging inspections used in previous revisions of this specification are as specified in Table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

Table X. Acceptance criteria in previous revisions.

Inspection	Paragraph	Acceptance Criteria in Previous Revisions
Appearance and Construction	4.4.2	AQL of 2.5% for major defects AQL of 6.5% for total defects
Dimensional	4.4.3	AQL of 1.5%
Packaging	4.4.4	AQL of 4.0%

6.7 Subject term (keyword) listing.

Cellular Rubber
Latex Foam Rubber
Rubber
Rubber Sheet
Rubber Shapes

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - MR
Navy - AS
Air Force - 11

Review activities:
Army - MI, AR, AL
Navy - MC
Air Force - 82
DLA - GS

User activities:
Navy - OS
Army - ME

Preparing activity:
Navy - AS

Project 9320-1094

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-5001B	2. DOCUMENT DATE (YYMMDD) 31JUL92

3. DOCUMENT TITLE Rubber Cellular Sheet, Molded and Hand-Built Shapes; Latex Foam

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDING OFFICER, NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION LAKEHURST SYSTEMS REQUIREMENTS DEPARTMENT	b. TELEPHONE (Include Area Code) (1) Commercial: (908) 323-7488 (2) AUTOVON: 624-7488
c. ADDRESS (Include Zip Code) CODE SR3 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:58Z
Check the source to verify that this is the current version before use.